Interview brief

# Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data, for the panel interview on Tuesday, November 3, 2026.


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Every figure comes from public data (EIA-861 and PUDL, NOAA Storm Events, EIA-930 and, with the upgrade, EAGLE-I county outages) or Dominion's own publications. No Dominion internal data was used. Material tagged **If adopted** depends on adopting the repo upgrade; the public-only line is given beside it. Items marked [you: ...] are for you to fill in.

The answer first

# A stormier year, a frequency gap, and storms by county

Dominion's reported outage minutes rose from 130 to 133; rebuilt from public EIA-861 data they are 130.2 and 133.0. 2025 had more storms and more interruptions, and once storms are counted there is no clear underlying trend. Against its peers, Dominion's gap is how often customers lose power, not how long.

**+2.8**

minutes: the rebuilt 3-year average, 130.2 → 133.0. 2025 (144.4) replaced 2022 (136.0) in the window

**+10%**

both storm county-days and interruptions per customer (SAIFI), 2022 → 2025; minutes per interruption (CAIDI) −4%

**4th highest**

of 12 on frequency (SAIFI 2021–25: 1.21); 6th of 12 on minutes per interruption. Peer medians differ by deck: page 4

## The eight-minute talk, slide by slide

| # | Slide | Say it in one line | Adopted · public |
|---|---|---|---|
| 1 | Title | Independent analysis of public data; nothing internal. | 0:35 · 0:32 |
| 2 | 130 → 133 | 2025 replaced 2022 in the window; SAIFI +10%, CAIDI −4%, storm county-days +10%. Dominion attributes the rise “primarily” to storms; 2025 sits on the storm line. | 1:11 · 1:11 |
| 3 | Dominion and 11 peers | Interruption length mid-pack (6th of 12); frequency 4th highest. Methods differ, so ranks are indicative. | 0:57 · 0:54 |
| 4 | Storms | If adopted County-days with a storm report had about 12 times the outage hours of calm days (11.9×, 95% CI 7.9–16.9): an association. Public-only: years with more storm days had more outage minutes (r = 0.70, ten years). | 1:04 · 0:57 |
| 5 | The Power BI report | One page per question, customer-weighted measures, a year slicer. Import for presentation (D13); If adopted through one folder parameter. | 0:54 · 0:57 |
| 6 | How it's built | RAW → STG → MART, tests, a data dictionary, a decisions log; reconciled to the reported 130 and 133. | 1:17 · 1:12 |
| 7 | First 90 days | The questions I'd bring and what I'd build; it ends on a Power BI program-tracking report. | 0:56 · 0:56 |
| 8 | Close | Three takeaways, then: “I've built a tested Snowflake → Power BI pipeline on public reliability data.” | 1:00 · 0:59 |

Times are each deck's speaker notes at about 130 words a minute: 7:54 in all (adopted), 7:38 (public-only).

## Say it carefully: the two-sided lines

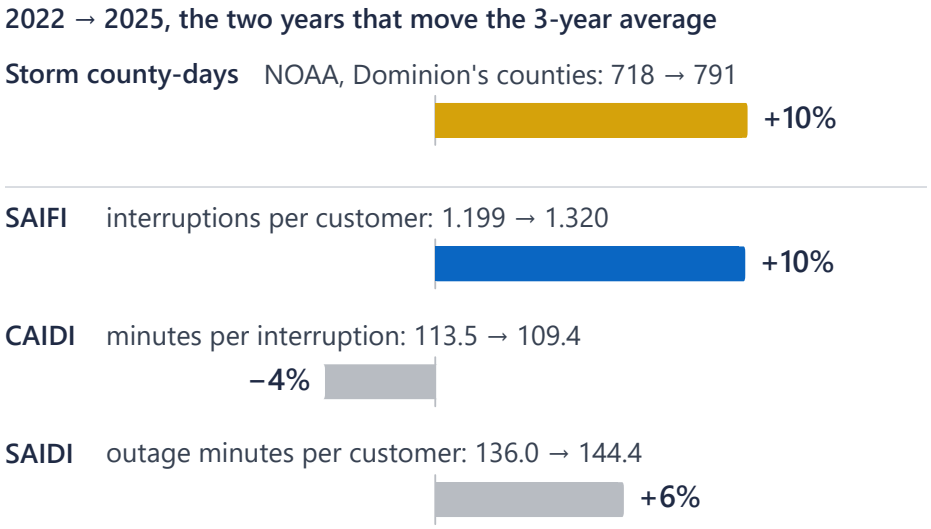
- Measured from 2023 instead of 2022, 79% of the change (in log terms) is frequency.
- 2025's SAIFI (1.320) only edges 2018's (1.303), and 2025's outage minutes (144.4) are below 2018 (149.6) and 2019 (149.9). One year is not a trend.
- No 2015–2025 trend is statistically clear. If adopted SAIFI p = 0.54, CAIDI p = 0.74, SAIDI p = 0.58 (scripts/trend_tests.py); public-only: say “no clear trend” without p-values.
- Adjusted for storm days, there is no clear trend since 2015. If adopted About +0.34 minutes a year, 95% CI −1.41 to +2.08, p = 0.67.

Finding 1

# Outage minutes rose from 130 to 133; 2025 had more storms and more interruptions

SAIDI, the average outage minutes per customer, is SAIFI (how often a customer loses power) times CAIDI (how long each interruption lasts). Dominion's 2025 Annual Report gives a 3-year SAIDI, excluding major events, of 133 minutes for 2023–25, up from 130 for 2022–24, and explains it (p. 10): "This increase is primarily due to increased storm activity." The public Virginia data reproduces both reported figures to the minute: 130.2 and 133.0.

A 3-year average moves when a year leaves the window: 2025 (144.4 minutes) replaced 2022 (136.0), a rise of 2.8 minutes in the average. Between those two years storm county-days rose 10%, SAIFI rose 10% and CAIDI fell 4%. The weather fit (Finding 3) puts 2025 on the storm line, so the public data agrees with Dominion's reading: once storms are counted there is no clear underlying trend.




Source: EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia (utility 19876), excluding major event days; NOAA Storm Events. Storm county-days 718 → 791 count Bedford city once (decision D22, If adopted); on the public repo they are 723 → 796, also +10%. Reported figures: Dominion Energy 2025 Annual Report, p. 10.

## The same split from either base year

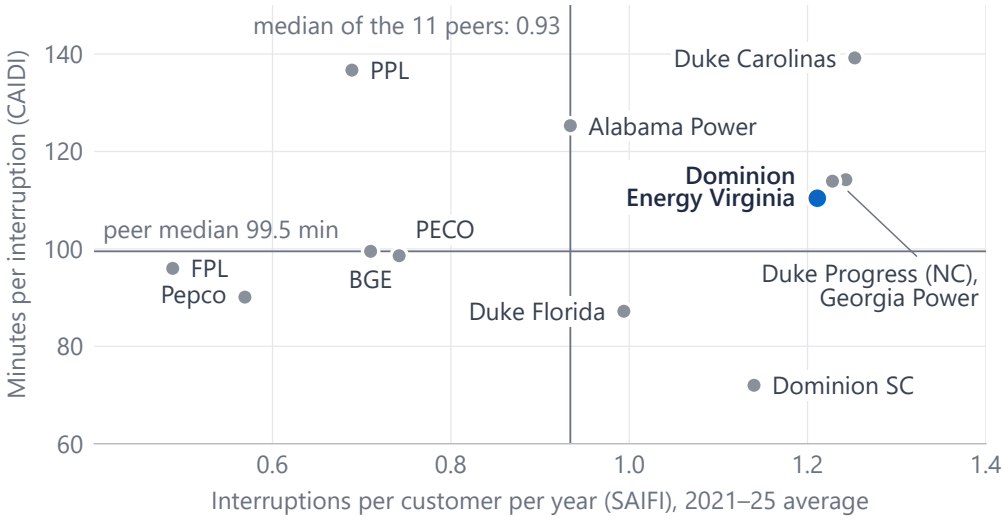
| Base year → 2025 | SAIDI | SAIFI | CAIDI | Share from frequency |
|---|---|---|---|---|
| 2022 (the year that left the window) | +6.1% | +10.1% | −3.6% | more than all of it |
| 2023 | +17.1% | +13.3% | +3.3% | 79% |

**If asked "so reliability got worse?"** "2025 had more storms and more interruptions than 2022, and Dominion attributes the rise primarily to storms. Once storms are counted there's no clear trend: 2025's SAIFI only edges 2018's, and its outage minutes are below 2018 and 2019. With outage records, I'd check whether non-storm interruptions are rising underneath."

Finding 2

# Among 12 large utilities (Dominion and 11 peers), interruption length is mid-pack; frequency is the gap

Twelve large investor-owned utilities in PJM and the Southeast, selected by EIA ID before results were computed (decision D10); means of 2021–2025, excluding major event days. Dominion's SAIFI of 1.21 is the 4th highest of the 12 (11-peer median 0.93); its CAIDI of 110 minutes is 6th, mid-pack (11-peer median 99.5); its 3-year SAIDI, 133.0, is 4th.




| Utility | Major-event method | SAIFI 2021–25 | CAIDI 2021–25 | SAIDI 2023–25 |
|---|---|---|---|---|
| Duke Carolinas (NC) | IEEE 1366 | 1.25 | 139.2 | 179.0 |
| Georgia Power | IEEE 1366 | 1.24 | 114.2 | 140.8 |
| Duke Progress (NC) | IEEE 1366 | 1.23 | 113.9 | 143.8 |
| Dominion Energy Virginia | Own method | 1.21 | 110.4 | 133.0 |
| Dominion Energy SC | IEEE 1366 | 1.14 | 72.0 | 83.9 |
| Duke Energy Florida | IEEE 1366 | 0.99 | 87.2 | 83.1 |
| Alabama Power | Own method | 0.93 | 125.3 | 110.0 |
| PECO | IEEE 1366 | 0.74 | 98.6 | 66.1 |
| BGE | IEEE 1366 | 0.71 | 99.5 | 64.2 |
| PPL Electric | IEEE 1366 | 0.69 | 136.7 | 99.1 |
| Pepco (MD) | IEEE 1366 | 0.57 | 90.1 | 48.3 |
| FPL | IEEE 1366 | 0.49 | 96.0 | 46.6 |

Source: EIA-861 / PUDL, excluding major event days, means of annual values. Georgia Power and Duke Energy Florida filed "Other" in 2013 only (public workbook: "Mixed"). Public-only deck: the 12-utility medians the public repo prints (docs/05), 0.96 and 105.

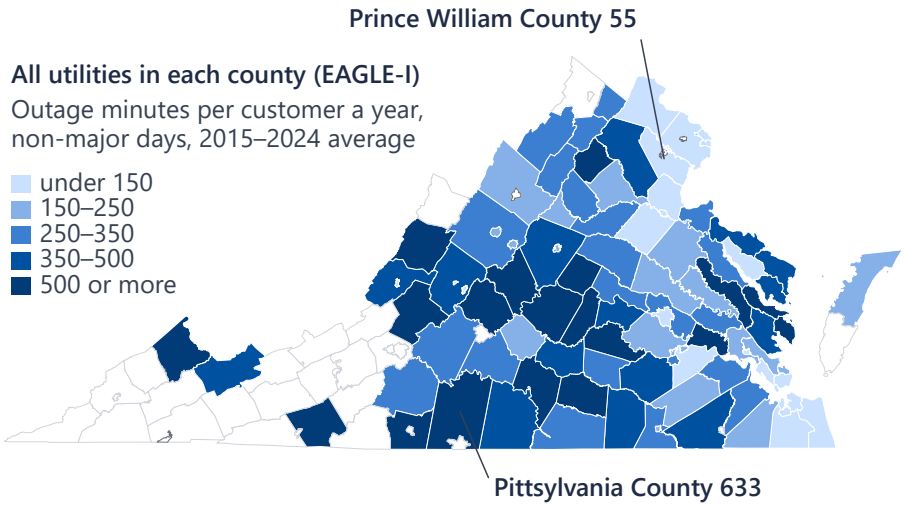
**The caveat to raise yourself:** methods differ. Dominion reports under its own major-event rule, so ranks are indicative ( If adopted 10 of the 12 file IEEE 1366 every year 2014–25; public-only, give no count). If asked: with all events, Dominion's SAIFI (1.531) ranks 6th of 12 (median of the other 11, 1.408), its SAIDI 2023–25 (223.0) 9th. Either way, the gap is frequency, not interruption length.

Finding 3

# Weather: years with more storm days had more outage minutes, and 2025 sits on the storm line

A storm county-day is a day with a NOAA wind, winter or tropical storm report in one of the 103 Virginia counties Dominion serves. Over 2015–2024, storm county-days and SAIDI excluding major events move together: r = 0.70 (n = 10). A fit made without 2025 gives 145.3 minutes for 2025's storm count, against 144.4 actual. Ten points is thin, so this is a check, not a forecast. If adopted p = 0.025; 95% CI for r 0.12 to 0.92; the 95% prediction interval for 2025 was 122.3 to 168.2 (the confidence interval of the line itself, 133.9 to 156.6). Storm county-days 2024 → 2025: 464 → 791 If adopted; 465 → 796 on the public repo.

> If adopted **County-days with a storm report had about 12 times the outage hours of calm days.** EAGLE-I, the Department of Energy's county outage record, counts the customers out every 15 minutes. On 6,108 storm county-days with complete outage readings (99 counties on the map), outage customer-hours per 1,000 customers averaged 332.8 (95% interval 225.6–453.4), against 27.9 (23.1–33.2) on calm county-days: 11.9 times (95% CI 7.9–16.9, resampling whole days). It holds on non-major days (9.9×), in the counties Dominion serves alone (19.9×) and in every year (5.9× to 27.4×, every interval above 1). It is an association: EAGLE-I counts every utility's customers, and storm reports partly follow damage. 6,108 is the NOAA count of 6,530 less 114 storm county-days in four counties off the map and 308 with a failed reading (96 of them in 2025, when only 74% of county-days are complete).



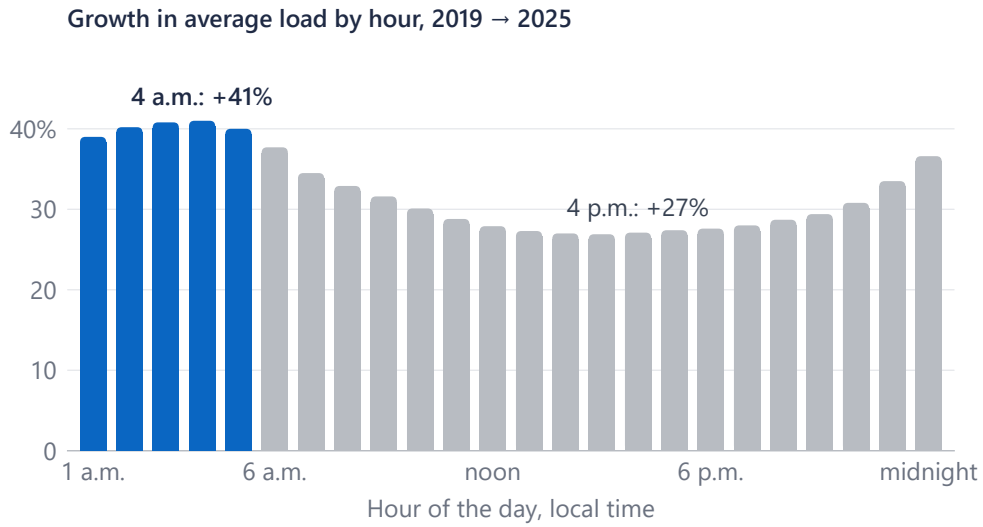

If adopted Source: EAGLE-I (ORNL for DOE), customers of all utilities in 99 of Dominion's 103 Virginia counties (four cities left off: unusable data). Counties with 20,000 or more customers range from 55 minutes a year (Prince William County) to 633 (Pittsylvania County). Not Dominion's SAIDI.

- If adopted The 31 mapped counties Dominion serves alone have a customer-weighted average of 159 minutes a year, against 211 in the 68 shared counties. That compares places (mostly cities and inner suburbs against rural counties), not utilities.
- If adopted Isaias, August 2020: 33.5 hours from the peak to clear 90% of the outage (351,738 customers out at the peak; a net count, all utilities, not a utility's "% of affected customers restored"). Across 47 events with 50,000 or more customers out, there is no evidence that clearing got faster (+4.6% a year after allowing for size, p = 0.07).
- What none of this can say: which circuits, which causes, or whether hardened lines held up. That needs outage records by circuit and cause.

Finding 4

# Overnight demand grew faster than afternoon demand, and winter rivals summer for the yearly peak

EIA-930 hourly demand for the PJM Dominion (DOM) zone, which is wider than Dominion's retail territory. From 2019 to 2025, average load at 4 a.m. rose 41%, against 27% at 4 p.m.; across the hours ending 1–5 a.m. the rise was 39–41%. The highest hour of the year rose 26% (19,803 → 24,957 MW) and the lowest 52% (7,167 → 10,903 MW). Winter set the yearly peak in 2019, 2022 and 2025.

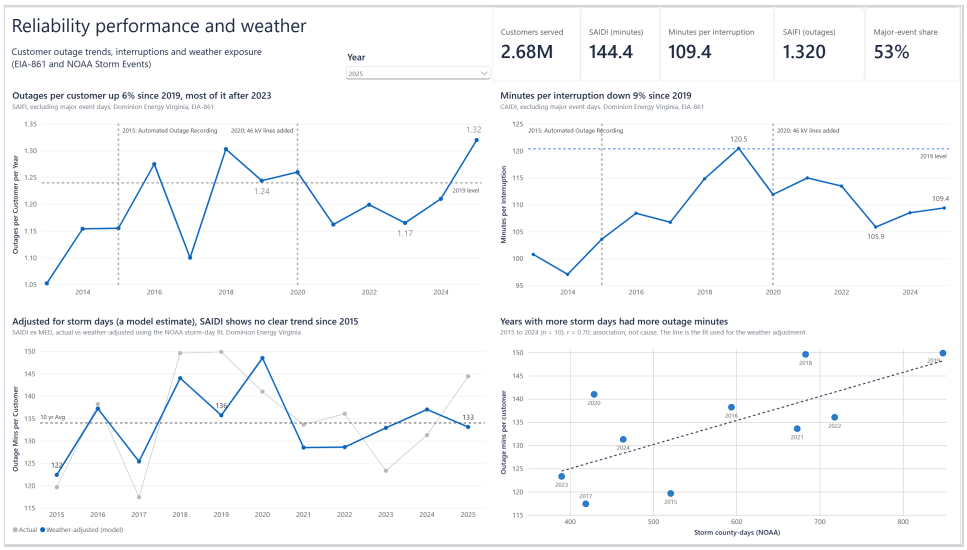



Source: EIA-930 hourly demand, PJM DOM zone; 2019 and 2025 full years. Winter = December–February, summer = June–August.

- Of the 50 highest hours in the EIA-930 record from July 2018 to December 2025, 47 fell in 2025.
- Commercial electricity sales in Dominion's Virginia territory rose 65% from 2013 to 2025 (39.08 → 64.62 TWh) on 8% more commercial customers (EIA-861).
- If adopted Holding Virginia's daily heating and cooling degree days and the weekday fixed, average load rose 33% (32% unadjusted), so weather does not explain it. 2026 to October 29: winter 25,413 MW against summer 25,384 MW, a tie.

**Don't say "data centers caused it".** Growth around the clock is consistent with large constant loads, but these data cannot identify customers. Say: "less overnight slack, and winter readiness matters as much as summer." In reports, I'd add winter and overnight views.

# The Power BI report, and how the analysis is built




If adopted The Summary page of the upgraded report, year 2025: customers served 2.68M, SAIDI 144.4, minutes per interruption 109.4, SAIFI 1.320, major-event share 53%. Public-only: deck slide 5 shows the frequency chart on your current report's What moved page. The .pbix is not on GitHub by design.

**The model.** One fact table, FCT_RELIABILITY (utility × state × year), many-to-one to DIM_UTILITY (names, peer groups); one page per question. Measures are customer-weighted: SAIDI is customer-minutes over customers within a year, and the 3-year figure is the mean of the three annual values, as Dominion reports it. The KPI cards follow a disconnected year table. Built in DirectQuery during development, Import for presentation (decision D13). If adopted 18 tables imported through one DataFolder parameter, so it opens offline; 11 pages, ending on Data notes (31 tests, 12 of 12 reconciled).

```
Selected SAIDI =
VAR SelectedYear = SELECTEDVALUE('Dashboard Year'[Year], 2025)
RETURN CALCULATE([SAIDI ex MED], FCT_RELIABILITY[YEAR] = SelectedYear)
```

**The build.** Public files land in Snowflake as raw text exactly as published (RAW), are typed and cleaned (STG), and become facts, dimensions and views (MART). Tests run after each load; all pass: T1–T6 the reliability mart, T7–T9 NOAA storms, T10–T12 EIA-930 demand ( If adopted T13–T31 the new tables). A data dictionary covers every table and column; measures are defined in docs/05. Decisions are logged (D1–D14; If adopted D1–D22); the peer set was chosen by EIA ID before results. Reconciled: 130.2 and 133.0 against 130 and 133, and test T4 matches 2025 to EIA's published 306.7, 144.4 and 2,675,735 customers ( If adopted 12 of 12 values: a check of definitions). scripts/run_local_duckdb.py rebuilds the core mart and runs its checks.

| Source | Publisher | Years | Rows |
|---|---|---|---|
| EIA-861 Reliability, 2025 final · PUDL EIA-861 | EIA · Catalyst Cooperative | 2025 · 2013–2024 | 977 · 24,358 |
| NOAA Storm Events | NOAA NCEI | 2013–2025 | 831,288 |
| EIA-930 hourly demand | EIA | Jul 2018–Oct 2026 | 6,077,232 · If adopted 6,094,320 |
| If adopted EAGLE-I county outages · DOE-417 · HURDAT2 · CPC degree days | ORNL for DOE · DOE · NOAA · NOAA | 2014–2025 · 2014–2025 · 1851–2025 · 2018–2026 | 11,608,442 · 3,702 · 55,524 · 309,504 |

# The role, and what I would do in the first 90 days

The posting that most likely matches this interview is **Associate Business Intelligence Analyst / Business Intelligence Analyst** (Richmond, hybrid: three days in the office, two remote). It supports **Grid Resiliency programs** with Power BI dashboards and reports, and asks for Power BI, SQL, Excel, Snowflake, Power Apps and SharePoint, with reliability metrics as preferred knowledge. [you: confirm the exact title and team with the recruiter]

- Four programs: mainfeeder hardening, voltage island mitigation, voltage optimization, and intelligent grid devices (fault location, isolation and restoration). The Strategic Underground Program is separate.
- Mainfeeder hardening selects feeders with at least 500 customers whose average SAIDI is more than 2 minutes worse than the rest, with above-average outages over the last five years. Dominion reports a 32% reduction in outage minutes on feeders hardened for at least 12 months (as of November 30, 2023).

## First 90 days: the questions I'd bring, and what I'd build

| Days | What I'd do | Deliverable |
|---|---|---|
| **1–30 · Learn, reconcile** | Learn how outages, meter events and circuits are recorded and reported. Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures. | A reconciled baseline with written definitions |
| **31–60 · Ask about frequency** | Interruptions by circuit and cause code, storm days apart from calm; repeat interruptions (CEMIn). Outage times from meter last-gasp and restore messages (99.8% of meters are smart), checked against OMS. | Tested views and a data dictionary |
| **61–90 · Track** | Track mainfeeder hardening against matched untreated feeders (same pre-period performance), weather-normalized, in support of the program's selection rule. | **A Power BI program-tracking report on a scheduled refresh** |

Check each of these first with the people who own the definitions: they are questions, not conclusions.

## What public data cannot show, and what would answer it

| Public data limit | What would answer it (Dominion data, by type) |
|---|---|
| One number per utility per year (EIA-861) | Outage records by event, circuit and cause |
| Storms counted by county, not by line | Outage records with cause codes and circuit location |
| Methods differ between utilities | Dominion's daily SAIDI and its filing method |
| The DOM zone is wider than the retail territory | Meter interval data by circuit |
| No customer identities: load growth cannot be attributed | Customer class and large-load records |

## Logistics

- Time, place and format: [you: …] · Panel names and roles: [you: …]
- Hybrid 3-2 in Richmond: [you: commute or relocation plan for a hybrid 3-2 schedule in Richmond] · Work authorization, if asked: [you: …]
- Salary, if asked: ask about the full scope first; give a researched range only if pressed. [you: your range]

Appendix A1

# Know every number

| Number | Plain meaning · how computed · source | Caveat → likely follow-up |
|---|---|---|
| **130 → 133** | Reported 3-year SAIDI ex-MED, 2022–24 → 2023–25 · Annual Report p. 10 | "Primarily due to increased storm activity" → is it real? (Q4) |
| **130.2 → 133.0** | The same, rebuilt: mean of three annual values · EIA-861 / PUDL | Same definitions, not proof the data is true (Q13) |
| **136.0 → 144.4** | SAIDI ex-MED 2022 (left the window) and 2025 (joined) · EIA-861 | (144.4 − 136.0) ÷ 3 = 2.8 minutes |
| **1.199 → 1.320 · 113.5 → 109.4** | SAIFI +10% and CAIDI (SAIDI ÷ SAIFI) −4%, 2022 → 2025 · EIA-861 | SAIFI only edges 2018 (1.303); CAIDI is a ratio, not restoration speed |
| **718 → 791 · 723 → 796** | Storm county-days 2022 → 2025, +10% ( **If adopted** · public) · NOAA | D22 counts Bedford city once |
| **145.3 vs 144.4 · r = 0.70** | 2025 from the storm-day fit (2015–2024, n = 10) vs actual; the fit's r | **If adopted**: 95% prediction interval 122.3–168.2; p = 0.025; CI for r 0.12–0.92 (Q1) |
| **79%** | 2023 → 2025 change: share from frequency (log split) | From 2022: more than all of it |
| **1.21 · 4th highest of 12 · 133.0** | SAIFI 2021–25 ex-MED; median **If adopted** 0.93 (11 peers), public-only 0.96 (all 12); SAIDI 2023–25, also 4th | All events: SAIFI 1.531, 6th (median 1.408); SAIDI 223.0, 9th |
| **110 · 6th of 12** | CAIDI 2021–25, mid-pack; median **If adopted** 99.5 (11 peers), public-only 105 (all 12) | The public repo prints the 12-utility medians (docs/05) |
| **10 of 12** | **If adopted**: file IEEE 1366 every year 2014–25. Public-only: "methods differ" (workbook: "Mixed") | Dominion, Alabama Power: "Other" → ranks indicative (Q2) |
| **11.9× (7.9–16.9) · 332.8 vs 27.9** | **If adopted**: outage customer-hours per 1,000 customers, storm vs calm county-days; the levels' intervals 225.6–453.4 and 23.1–33.2 | An association; EAGLE-I counts all utilities (Q3, Q5) |
| **9.9× · 19.9× · 5.9–27.4×** | **If adopted**: non-major days · Dominion-only counties · each year | Every year's interval is above 1 |
| **6,108 vs 6,530** | **If adopted**: storm county-days with complete readings (99 counties) vs all (103) | 422 = 114 off the map + 308 failed (96 in 2025) |
| **306.7 vs 144.4** | 2025 SAIDI with and without major event days; MEDs 53% (162.3) | Dominion vs FPL: compare like with like (Q2) |
| **2.68M** | Customers served, 2025: 2,675,735 · EIA-861 | One SAIDI minute ≈ 2.7M customer-minutes |
| **T1–T12 · D1–D14** | Tests, all pass (T1–T6 reliability, T7–T9 storms, T10–T12 demand); decisions logged. **If adopted**: T13–T31 on the new tables (31 in all), D1–D22, 12 of 12 reconciled | Reconciliation is a definitions check (Q13) |
| **Rows** | EIA-861 977 + 24,358 · NOAA 831,288 · EIA-930 6,077,232 ( **If adopted** 6,094,320) | **If adopted**: EAGLE-I 11,608,442; 18 tables, 11 pages |
| **99.8%** | Smart meters as a share of all meters, 2025 · EIA-861 | A last gasp marks the start (Q6) |
| **500 · 2 min · 5 yr · −32% · −44%** | Mainfeeder selection rule. Reported: −32% outage minutes, feeders hardened 12+ months (Nov 2023); −44% average annual outage time, hardened areas (June 2026) | Two measures, not a trend; −44% has no stated baseline. Before/after overstates it (Q8) |
| **+41% · +27% · +26% · +52%** | Load 2019 → 2025: 4 a.m., 4 p.m., top hour, lowest hour · EIA-930 | Not attributable to data centers (Q10) |
| **1.849 → 1.887** | Co-op SAIFI 2025 before and after matching denominators (P18) | A co-op filed SAIDI but no SAIFI (Q17) |
| **50 basis points** | How far the SCC can move the allowed return (factors include reliability) | Why the numbers have to be right |
| **20M+ · 71% → 90% · 17% · +9%** | Mercury: rows; grounded-answer accuracy; settlement time; renewals | Causal only for the A/B test (Q17) |
| **$690M · 440** | The Life Initiative: investments in the board's Tableau view | Who used it? [you: …] |
| **25% · 3-2** | Travel up to 25%; three office days, two remote · the posting | [you: your answers] |
| **May 2026 → H2 2027** | NextEra merger announced; Virginia order by mid-January 2027; close expected | No view on the deal (Q23) |

Appendix A2–A3

# The 25 hardest questions, by panelist

**C** = claim, **P** = proof, **L** = honest limit. ● = drill these first. *Adopted* = depends on adopting the repo upgrade; *Public-only* = the fallback line. [you: …] = only you can fill it. D = decision, P18 and the like = data-quality notes, T = tests (repo docs). Q numbers match the "likely follow-up" column in A1.

## Engineer

**Q1 ● Weather fit.** *"Ten points: what's the interval on r, the 2025 band, and why start in 2015?"*
**C:** Thin, so it's an appendix check, not the headline. **P:** r = 0.70, p = 0.025, 95% CI for r 0.12–0.92. · 2025: fit 145.3 vs actual 144.4, inside its 95% prediction interval, about 122–168. · The fit starts in 2015 because that's when outage recording became automated (D6). **L:** Say "can't tell", not "no trend". *Adopted:* the weather-adjusted slope range is −1.4 to +2.1 min/yr; storm county-days 464 → 791. · *Public-only:* r 0.70 on ten years; 465 → 796; no p, CI or prediction interval.

**Q2 ● Major event days.** *"What's an MED, and why exclude it? What's Dominion's rule? Dominion vs FPL?"*
**C:** An MED is a day so far above normal that it would swamp the year. Excluding MEDs shows everyday performance; storms are judged as storm response. **P: 2.5β:** the logs of five years of daily SAIDI; the threshold is T = e^(α+2.5β). · **2025:** MEDs were 53% of all-event minutes (306.7 vs 144.4). · **Dominion's method:** it files under its own rule ("Other"), so methods differ. *Adopted:* 10 of the 12 use IEEE. · **All events:** Dominion's SAIFI ranks 6th of 12, against 4th ex-MED. **L:** "I don't know the written rule; I'd ask the reliability team." Ranks are indicative. FPL: compare like with like (all events) and make no comment on the merger.

**Q3 ● County data (*Adopted*).** *"It counts every utility in a county. Whose minutes? Would you send crews by it? 90% of what?"*
**C:** Right. It isn't Dominion's SAIDI, and I wouldn't target work with it. **P: The county rule:** EIA-861 lists 103 Dominion counties, 33 of them Dominion-only (D15), 31 on the map. The result holds in those 31 (19.9×). · **Blank readings:** failed collection runs, never counted as zero (P20). · **Isaias:** 33.5 hours from the peak to clear 90% of the outage (a net count, all utilities). **L:** Targeting needs your outage records by feeder. *Public-only:* there's no county outage data; NOAA's county reports carry the same caveat.

**Q4 Anything to explain?** *"130 → 133 is 2.8 minutes. Is it real? More frequent, or longer?"*
**C:** 2025 was a stormier year with more interruptions. There's no clear underlying trend. **P: The 2.8 minutes:** (144.4 − 136.0) ÷ 3, because 2025 replaced 2022 in the window. Single years have ranged from 117 to 150. · **2022 → 2025:** SAIFI +10%, CAIDI −4%; storm county-days 718 → 791. · **Dominion's own reading:** "primarily due to increased storm activity". **L:** CAIDI is a ratio, not a measure of restoration speed. *Adopted:* trend p-values 0.54 / 0.74 / 0.58. · *Public-only:* storm county-days 723 → 796; "no clear trend".

**Q5 The 12× (*Adopted*).** *"Isn't 12× circular, when storm reports come from damage? What's the unit of the CI?"*
**C:** Partly, yes. It's an association, inflated by how storm reports are made. **P: The levels:** 333 vs 27.9 customer-hours per 1,000 customers = 11.9× (7.9–16.9). · **The interval:** bootstrapped by whole days. · **It holds:** on non-major days (9.9×), in Dominion-only counties (19.9×) and in every year. · **6,108 vs 6,530:** complete EAGLE-I days in 99 counties vs all NOAA days in 103. **L:** The next test is weather measured independently of damage, such as wind gusts. *Public-only:* the annual version, r 0.70.

**Q6 Meters.** *"What does a last gasp tell you? Could 2025's SAIFI be a measurement change?"*
**C:** A last gasp marks when the outage starts, not when it ends. I can't rule out a measurement effect from public data. **P:** 99.8% of meters are AMI. · The end time needs a restore message or a ping, and the outage system stays the record. · Last gasps drop out when collectors lose power. · Recording became automated in 2015 (D6). **L:** Check first: roll-out timing against SAIFI, and circuits before and after their meter cut-over.

**Q7 Cause codes.** *"What will our cause codes show, vegetation especially? What goes wrong?"*
**C:** It needs your outage records: a 30–60-day deliverable. **P:** Expected causes: vegetation, equipment, weather, animals, vehicles, unknown. · Split blue-sky days from storm days, and use CEMIn for repeat outages. · Dominion trims on 4–5-year cycles. **L:** The data traps: the cause is coded when the job closes · one device can hit many customers · the "unknown" share is itself a KPI · Agree the definitions with engineering first.

**Q8 Hardening.** *"The worst feeders improve anyway. How would you know it's the program?"*
**C:** A plain before-and-after overstates the effect (regression to the mean). **P: Selection rule:** at least 500 customers, more than 2 minutes worse, over 5 years. · **Reported, two measures:** −32% outage minutes, feeders hardened 12+ months (Nov 30, 2023); −44% average annual outage time for customers in hardened areas (Dominion, June 2026). · **Design:** matched feeders with a pre-period, difference-in-differences, storm-normalized, reported with intervals. · **My experience:** the A/B test at Mercury. **L:** Frame it as making a good result harder to challenge, not as doubting it.

**Q9 ● Why EIA.** *"Why public EIA data and not the SCC filings?"*
**C:** It's public, consistent across utilities, and it reproduces Dominion's own numbers. **P:** Every utility files the same federal form, by state. · Virginia data gives 130.22 / 133.01 against the reported 130 / 133. · The SCC's report to the Assembly shows the indices only as charts. **L:** Methods differ (Dominion files as "Other"), so ranks are indicative. The filed SCC report is the first thing I'd reconcile to.

**Q10 ● Data centers.** *"Did data centers cause the demand growth?"*
**C:** Public data can't show that, and I don't claim it. **P: The load shape, 2019 → 2025:** overnight +39–41% vs afternoon +27%; winter peaks. · **Sales:** commercial sales +65% on 8% more customers. · **Dominion's account:** 744 MW of data centers connected in 2025, and over 54 GW in contracting stages. **L:** The zone is wider than the territory and EIA has no customer IDs. Attribution needs meter data (D14).

## BI manager

**Q11 ● The model.** *"Walk me through the model and the year slicer, 3-year SAIDI and the roll-up."*
**C:** A small star schema; every number comes from a customer-weighted measure. **P: Tables:** FCT_RELIABILITY (utility × state × year) → DIM_UTILITY. · **Year slicer:** a disconnected 'Dashboard Year' table read with SELECTEDVALUE (default 2025) plus CALCULATE. · **3-year SAIDI:** the mean of the annual values (AVERAGEX), Dominion's convention. · **Roll-up:** Σ customer-minutes ÷ Σ customers. **L:** DirectQuery while developing, Import to present (D13). *Adopted, after the guide:* folder-parameter Import, so it opens offline.

**Q12 Power BI at work.** *"Where have you used Power BI at work? And Power Apps?"*
**C:** [you: the exact truth] . If it's only this project, say so. **P: At work:** Tableau; Snowflake SQL. · **Power BI depth:** this model (Q11). **L: Power Apps:** "not in production" [you: ...] , plus a 30-day plan (an exceptions form). · **SharePoint:** skills line only [you: ...] .

**Q13 Reconciliation.** *"133.0 vs 133: what did that prove? Which values?"*
**C:** It shows I use the same definitions and filters as Dominion. It doesn't show the data is true. **P:** 12 values: two 3-year averages vs the annual report · nine 2024–25 values vs EIA's files · 2025 commercial sales · It confirms the row, the MED flag, the averaging, the vintage and the customer base. Tests run after every load. **L:** It's a definitions check. A real reconciliation runs from outage events to daily SAIDI to the filed figure, with a written bridge. *Public-only:* 130.2 / 133.0 vs 130 / 133; tests T1–T12.

**Q14 Governance and regulators.** *"What does 'governed' mean? Any regulatory reporting? Why doesn't your dashboard SAIDI match the filed number?"*
**C: Experience:** [you: the truth; no job bullet shows a filing] . · **Governance:** owned definitions, change control, and checks that block bad data. **P: What it takes:** a versioned glossary, sign-off, lineage, access rules, alerts. · **What I've done:** definitions written up for reuse; the board's dashboard. · **Filed vs dashboard:** the filed number anchors. Write the bridge (population, MED rule, customers, vintage) and keep as-filed and revised versions side by side. **L:** The project has a decisions log, a dictionary and tests, but no owner or monitoring. Say so.

**Q15 EAGLE-I in SQL (***Adopted***).** *"How do 15-minute snapshots become events and customer-hours? What about missing data?"*
**C:** It's in the same SQL build, under tests T13–T31. **P: Customer-hours:** Σ customers out × 0.25. · **Events:** gaps-and-islands (LAG, then a running SUM). · **"Held 1 hour":** the minimum of four readings. · **Time:** Eastern Standard Time all year. · **Bad data:** blanks are NULL, not zero (P20); impossible readings are NULL (P24). · **Size:** 11.6M rows. **L:** Coverage grows over time, so compare a county with itself. There's no stuck-count rule yet. *Public-only:* describe it as how you would do it.

**Q16 Mercury pipeline.** *"The 20M-row Snowflake pipeline: grain, loads, and which part was yours?"*
**C:** [you: what it fed and what you owned] . **P:** Policy, claims and billing data (20M+ rows) for the pilot. · [you: the grain, incremental logic, scheduling, tests, one failure] **L:** Claim only your part. Bridge: outage events → daily → annual indices.

**Q17 Real reconciliation.** *"That's model evaluation. Give me a real reconciliation. Were the sales lift and the settlement cut yours?"*
**C:** Evaluation and reconciliation are different, and I keep them apart. **P: Evaluation:** a 150-question test set, 71% → 90%. · **At work:** [you: one true break → cause → control] . · **In the project (P18):** a co-op with SAIDI but no SAIFI would have diluted the group figure (1.849 vs 1.887). Fixed Oct 27. **L:** Causal only where there was a control: the A/B test, +9%. Elsewhere, "contributed to".

**Q18 ● Do differently.** *"What would you do differently?"*
**C:** Set rules before seeing results, as I did for the peer set (D10). **P:** Test aggregates for missing values first (P18). · Show the all-events view beside ex-MED: the rank moves from 4th to 6th. · Set one stated rule for windows and base years. **L:** [you: one personal lesson] . Never tell an upgrade change as a discovery story.

## HR

**Q19 Why move.** *"Why leave Mercury after seven months? Why a utility?"*
**C:** [you: the real reason, as a pull] . One neutral line on Mercury. **P: The Grid Resiliency programs:** Mainfeeder Hardening, Voltage Island Mitigation, Voltage Optimization, Intelligent Grid Devices/FLISR. · [you: which one, and why] · **My project:** rebuilt Dominion's numbers. **L:** Never criticize Mercury.

**Q20 Three years.** *"'Three years of BI', but your data titles add up to under a year. And the Associate level?"*
**C:** Here's how it adds up: **Data titles:** about 10 months (Mercury 7, Life Initiative 3). · **Analytics inside other roles:** about 28 (Cofoods BD 19, with the forecasting model and Tableau dashboards; finance internships about 9). · **Total:** about 38 months since April 2022, plus the MS. **L:** "Three years" counts all work, internships included; don't stretch it. Open to either level; earn the step up.

**Q21 Pushback.** *"Someone senior said your numbers were wrong. And our engineer's pushback?"*
**C:** Listen, check, fix the source, document it, thank them. **P:** [you: a true STAR story; the resume has none] · **Live:** "You're right, it counts every utility. I'd relabel it or cut it to Dominion-only counties." (Q3) **L:** Experts own the definitions; the analyst owns reproducibility.

**Q22 Least sure.** *"Which number are you least sure of? And what do you say when you don't know?"*
**C:** Name one, and say why. **P: Candidates:** the weather fit (ten points); Dominion's MED rule (Q2); *adopted:* the county minutes (Q3). · **"I don't know":** "Here's where I'd find out, and I'll come back by [time]." [you: a time you did this] **L:** Doubt builds credibility only when the fix comes in the same breath.

**Q23 Merger.** *"How do you feel about joining during the NextEra merger?"*
**C:** Calm. Reliability work carries on, and integrations need exactly this definition work. **P: Timeline:** Virginia's order is due by mid-January 2027; SC, NC, FERC and NRC approvals follow; close is expected in H2 2027. · **The company says** Dominion Energy Virginia stays separately regulated. **L:** No view on the deal or on jobs, and no comment on FPL. Ask how priorities may shift.

**Q24 Thirty seconds.** *"No acronyms: what did you find?"*
**Say:** "In 2025 the average Dominion customer in Virginia lost power about 1.3 times, for about an hour and fifty minutes each time: almost two and a half hours in the year, leaving out a few extreme storm days. The three-year average rose about three minutes, in a stormier year. Once storms are counted, there's no clear trend. This job is about showing clearly whether the programs that prevent outages are working." **L:** Stop at thirty seconds.

**Q25 Unglamorous work.** *"How do you recommend things without overstepping? Are you happy doing routine reports?"*
**C:** Yes. Accurate recurring reports are the job. **P: Habits:** write-ups for reuse; handover notes; the project's dictionary and decisions log. · **Raising ideas:** as questions, with the data, checked with the owners, and credit to them. **L:** Learn the planning context before recommending.

Appendix A4

# Delivery and rehearsal

Strength: **strong** = meta-analysis or large review; **single** = one study; **mixed** = replications disagree; **practice** = common advice with no study verified (kept and labeled, never cited).

| Tip | Rests on | Strength | Do this |
|---|---|---|---|
| **The first minutes count** | Barrick, Swider & Stewart 2010 | single | Open with a crisp 30-second answer on what you do and built; early impressions tracked final ratings (r = .42). |
| **Sell with facts; never stretch** | Swider et al. 2011; Barrick et al. 2009 | single + strong | Real results helped (r = .20); "image creation" hurt (r = −.11 to −.19). Use the resume's numbers exactly. |
| **Expect past-behavior questions** | Sackett et al. 2022; Taylor & Small 2002 | strong | Every answer is scored on a scale. Have 5–6 true STAR stories: situation, *your* action, result. |
| **State the limit, then answer it** | O'Keefe 1999 | strong | A limit builds credibility only with its answer: "ten points is thin, so the headline uses county-days". |
| **Ask a follow-up question** | Huang et al. 2017 | moderate (conversations, not interviews) | At "Any questions?", first follow up on something a panelist said. |
| **Rehearse aloud** | Powell, Stanley & Brown 2018 | strong | Anxiety and scores correlate at r = −.19. Rehearse to a mock panel; that it calms you is a reading, not a result. |
| **Call the nerves "excitement"** | Brooks 2014; Bosshard & Gomez 2024; Ganley et al. 2021 | mixed, small | Say "I'm excited", not "calm down": the pooled effect is small (d ≈ 0.2) and a math replication found nothing. |
| **Answer first** · **"I don't know" → how I'd find out** · **Pace, close, thank** | none verified | practice | Claim, one proof, then the limit. Never guess a number; say where you'd look and when you'll come back. Pause after each answer, close in one line, send a thank-you note within a day. |
| ~~Power posing~~ | Ranehill et al. 2015 | failed | Left out. |

## Three interviewers

Panelists may score separately and then combine; the panel research is mixed (Dixon et al. 2002), so these rules are inference. Make each answer complete on its own. Answer the asker, then turn to the others for your last sentence. Take a challenge with thanks: concede what's right, then answer.

| Panelist | Listens for | Lead with | Questions |
|---|---|---|---|
| **BI manager** | How it's built; checked numbers; Power BI | The model, the measures, the reconciliation; the printed report page | Q11–Q18 |
| **Engineer** | Exact definitions; honest limits | Precise terms, a fast concession, the data that would settle it | Q1–Q10 |
| **HR** | Plain words, fit, how you take pushback | No acronyms (Q24), true STAR stories, calm on the merger | Q19–Q25 |

## Rehearsal outline for Monday

1. Decide the variant first (adopted or public-only); everything after depends on it.
2. Apply the Power BI guide for your variant; print one report page and the model view for Q11.
3. Read this brief once, for understanding; mark anything you couldn't explain.
4. Say the deck aloud against an 8-minute timer, three times; record one run. Cut words, not slides.
5. Drill the question sheet and the 25 hardest by recall: cover the answer, say it aloud, then check (Roediger & Karpicke 2006; Rowland 2014: strong).
6. Spread the drills: an afternoon round and an evening round of the misses (Cepeda et al. 2006: strong).
7. Stop at 20:00 and sleep (Lim & Dinges 2010: strong). Tuesday: say the 30-second opener twice and leave early.